Wireless InSite Change Log

3.4.4.13

Bug Fixes

- Upgraded OptiX library to 6.0.0 to support newer GPUs.

3.4.4.12

New Features

- Added creation of walls for a floor plan when creating it using the outline of a substructure.

Bug Fixes

- Fix bug where mobile platform simulations could hang in after processing geometry due to a study area thread count exceeding app preferences limit
- Fixed a bug which prevented the export of H-matrix files when a transceiver was selected.
- Fixed an issue where TxRx sets not being edited would render out of alignment in the editing window.
- Fixed an issue where modal pop up windows would be hidden behind parts of the UI. Fixed an issue where pop up windows would not appear on the expected display.
- Fixed an issue where Tx base stations would intermittently be removed from comm systems.
- Create EES plate option now disabled when there is not an active project.
- Fixed an issue where Wireless InSite would crash if a user selected a door or window before making a wall.

3.4.4.11

New Features

- Added ability to associate network parameters defined in a Touchstone file with a MIMO antenna to describe impedance mismatch conditions and mutual coupling between elements.

Bug Fixes

- Resolved bug when indexing antenna parameters to generate binary antenna outputs for scenarios involving MIMO antennas and multiple instances.
- Resolved bug where binary output antenna data was not being stored based on correct frequency for receiver side.
- Materials can now be added to a Monte Carlo multi-material.
- Monte Carlo multi-materials will now correctly read in their ID so that faces can reference them for rendering and property display.
- Locating an S-parameter file for a MIMO antenna will ensure the file is available in the project folder when specified through all work flows.

3.4.4.10

New Features

- Improved error handling for MPI simulations.
- Added ability to specify the incident angle to the surface normal when generating an EES material.
- Allow duplication of surface illumination transceiver sets.

Bug Fixes

- Fixed an issue where comm systems weren't properly accounting for the user's power unit preferences.
- Default to using solid line style for plots when adding them to existing graphs through the Multi-Channel Results Browser.
- Exiting the MIMO properties dialog will no longer cause a crash when using RDP.
- Fix failures to read/parse user-defined material files.
- Fix failure to warn user when user-define material files have invalid entries.
- Importing plots from *.plt files will copy the file into the project folder so that they can be reloaded.
- Moved analysis type field in communication system properties window to make it stand out.
- Updated the geometry view's menus to match those in the main window.
- Fixed an issue where MIMO Tx wouldn't allow the Total Array Power toggle to be used.

3.4.4.9

New Features

- All analytic antenna types now support multiple frequencies.

- Added a scale bar mode to allow Throughput values to be shown precisely. Added a throughput unit type to allow users more granular control over viewing output in the UI.
- Added advanced partitioning and queuing parameters exposed by an application preference.
- Added new viewing mode to only display geometry that is selected.
- Added control of SU-MIMO data streams to optionally 1) optimize power allocation, 2) set stream number limit, 3) force equal MCS across streams.
- Updated the progress reporter to show progress while importing an indexed antenna.
- Added an option for users to specify if plot styles should be updated automatically for readability when being added to an existing graph.
- New simplified frequency sweep behavior in GUI interface and API.
- Power sum results for diffuse scattering are now power summed at both subfacet and path levels.
- Changed the default behavior to not modify the drawing style when adding a new plot to an existing graph.
- Added a tool to generate EES material files and assign them to either a new plate feature, a face that is positioned with respect to another with a specified offset, or directly to a selected face.
- Added ability to select the frequency to use for viewing the antenna pattern of the selected element in the MIMO properties dialog.
- Added RX Codebook functionality to communication systems.
- Extended built-in throughput tables to include 5G-NR and 802.11ax.

Bug Fixes

- Real time models will no longer launch when an active feature has an EES material assigned to a face.
- Calcprop models will no longer launch when an active feature has an EES material assigned to a face.
- The display of results from a multi-column result file can now be toggled off and back on.
- Seeded random number generator to keep duplicate facet removal (and results) consistent.
- Fixed an issue that caused X3D simulations with building penetration receiver sets to crash prior to running.
- Fixed an issue that caused X3D simulations with Monte Carlo multi-materials to crash prior to running.
- Replacing/Deleting EES materials now removes the EES material from the material list.
- Fixed an issue where the study area entry in the project tree would disappear after running a simulation.
- Replacing/Deleting EES materials now removes the EES material from the material list.
- Prevent editing operations on faces with EES materials attached that would cause the reference to become invalid.
- Replacing/Deleting EES materials now removes the EES material from the material list.
- Fixed an issue with processing long lines of text in geometry feature files.

- Triangulating a feature with ees materials on it is no longer allowed.
- Change height now does scale by in the z axis with an offset to do the same function as change height while keeping ees materials attached.
- Translate in geometry view before and after triangulation with ees material works as intended.
- Triangulating is not allowed for objects with ees materials.
- On stream in there is a check for ees materials to be set to multiple faces and if there is make a copy for the duplicate faces.
- Simplify for all geometry has been removed as an option.
- Engineered surface transmissions now count towards diffuse scattering interaction limits.
- Fixed an issue where the TxRx display point size was not correctly changing the rendered size in the geometry view.
- Corrected missing diffuse scattering paths when APG enabled.
- A unique description is assigned to new multi-frequency user-defined antenna when added to a MIMO antenna. This is required in order to maintain encapsulation of the antennas supporting files with respect to other contents of the project.
- Improved EES diffraction calculations.
- Fixed an issue where some indexed antennas couldn't be used properly on Linux.
- Fixed an issue where the maximum number of streams in a communication system could not be returned to the automatic setting.
- Fixed an issue where the degree symbol was not being displayed properly.
- Only allow selection of waveform type when creating it through the MIMO antenna editor if calcprop options are exposed through the preferences.
- Corrected diffuse scatter path ray leaks by adding partial path check for front/back DS hits.
- Ensure that the origin of an EES material remains in the plane of the face that it is assigned to.
- Decreased EES diffraction cone spacing for some cases involving diffusers.
- Removed EngCompD paths for non-knife edges.
- Correctly apply distance from transmitter in calculating ray density for EES Transmissions.
- Fixed an issue where multi-frequency user defined antennas encountered a waveform error when being used as a Tx or Rx.
- Multi-Channel results browser is now available for general use. MIMO license checks remain in effect when invoking an action on a selected channel.
- Changed the organization of the application preferences documentation and the text of the default application preferences tab in the ui.
- Fixed an issue where multi-frequency user defined antennas encountered a waveform error when being used as a Tx or Rx.

3.4.4.8

New Features

- Added support for multi-frequency user-defined antennas as elements in MIMO antennas.
- Added transmissions through Electrically Engineered Surfaces (EES) that allow them.
- Added new type of Electrically Engineered Surface (EES) using Modal Impedances.
- Optimized ray-tracing time for diffusing Electrically Engineered Surfaces (EES).
- Added user-specifiable interaction limit for reflections and transmissions on diffusing Electrically Engineered Surfaces (EES).

Bug Fixes

- Adding a pattern to a multi-frequency user-defined antenna requires specifying a frequency so it does not overwrite an existing one.
- Added a check for valid VSWR values for each entry in a multi-frequency user-defined antenna.
- Adding a user-defined antenna to a MIMO antenna will only bring up the file chooser dialog once.
- Removing a multi-frequency user-defined file from a MIMO removes its underlying folder and pattern files.
- Projects with MIMO antennas that used a master index file for its multi-frequency user-defined antennas will open without error.
- Removing a MIMO antenna with multi-frequency user-defined antennas removes their underlying folder and pattern files.
- New waveforms no longer overwrite existing ones when added to the project.
- Context menu wording change for more clear option descriptions.
- Choose Material Window made wider for less horizontal scrolling.
- Edit array option is properly hidden by calcprop toggle.
- Tooltip added in comm system properties window for the ability to view the full file path.
- Plot legend position range restricted correctly, and size minimized whitespace while always maintaining enough width for text.

3.4.4.6

Bug Fixes

- Correct DS path setup for exact path and path completion logic bug causing non-repeatable output

3.4.4.5

Bug Fixes

- Mimo license is no longer needed to run Monte Carlo projects.
- The power per stream for singular value decomposition is now calculated correctly when the number of Tx elements exceeds the number of streams.

3.4.4.2

Bug Fixes

- Corrected issue with display of stale results after saving project to a new location and running.
- Fixed an issue where Tx/Rx origin properties could not be saved.
- Engineered surface materials will now position correctly when attached to faces in features that are relative to terrain.
- Fixed an issue that caused engineered surface materials to be incorrectly adjusted when relative to terrain.
- Previous versions are now forward compatible with new sorting levels of the tree. This was causing nodes to appear without a description.
- All results are added to the tree when 'Data Type' is selected as the last sorting level in the general application preferences.

3.4.3.0

New Features

- Illegal Study Areas are checked and alerted to the user based on Windows and Linux directory name restrictions.
- Added ability to sweep UAN patterns with frequency sweeps.
- Added a command line option to the CIR export utility to enable specifying an output location.
- H-matrix outputs are now formatted to agree with the standard definition found in SU-MIMO literature (which is the conjugate transpose of previous format).
- Antennas can now be copied into a MIMO antenna from the project's main list.
- Externally-queued simulations no longer require a job partitioning method.
- Adding support for user-registered antennas in the Wireless InSite API.

Bug Fixes

- Transmitter and receiver flags can now be deactivated in a properties session after being activated.
- All points in a surface illumination set will now be displayed.
- X3D simulations using EQI no longer write intermediate sub-job XML files to disk. This allows for multiple EQI study areas in a single project to run in parallel.
- Fixed an issue where attempting to load a default object material could cause a crash.
- Fixed an issue where both Monte Carlo material types did not require a short description, leading to loading and crashing errors.
- Fixed an issue where double clicking a setup file would launch the appropriate software, but fail to load the project.
- Fixed an issue that prevented horizontal paths from being produced by the Triple Path Geodesic model.
- Fixed an issue that caused the VPUP and Triple Path Geodesic models to sometimes crash if a receiver point was located in or on a facet.
- Renaming multi-frequency antennas is adjusted to ensure that its underlying folder does not collide with pre-existing ones.
- Deleting a multi-frequency antenna will remove its files.
- Manually created multi-frequency antennas will now put files in expected location.
- Added multi-frequency antenna importer warning for bad file references.
- Simplified the antenna import menu.
- Fixed an issue where study area selection and highlighting was inconsistent.
- Transmitter and receivers configured to use the waveform assigned to their antenna will now resolve correctly.
- Transceiver sets are now set to use the antenna's waveform when created.
- Waveforms with an invalid project ID can no longer be added to a project.
- Creating a new waveform through the transmitter or receiver editing dialog will automatically create a Sinusoid waveform when calcprop options are not exposed.
- Transceivers table refreshes when an antenna's waveform is changed.
- Fixed an issue where attempting to load an invalid project would leave the application title bar empty.
- Fixed an issue that prevented horizontal paths from being produced by the Triple Path Geodesic model.
- Fixed an issue that caused the VPUP and Triple Path Geodesic models to sometimes crash if a receiver point was located in or on a facet.
- Renaming multi-frequency antennas is adjusted to ensure that its underlying folder does not collide with pre-existing ones.
- MIMO output browser indexing synced with p2m output, starting with 1 instead of 0.

3.4.2.0

New Features

- New Surface Illumination Transmitter/Receiver set type. This provides the ability to cover building exteriors, roof tops, and insert receivers at regular floor heights. It replaces the implementation of the *Cover with points* context menu option for selected geometry.
- Adding the option to write path id files for X3D simulations. (Available via API only.)
- Adding CIR export tool to the utilities distribution.

Bug Fixes

- Context menu now appears in correct location if invoked on the non-primary monitor.
- Changing height in Floorplan editor while in unsaved project no longer produces unwanted walls, floors, and ceiling behavior.

3.4.1.0

New Features

- Roughness correction now uses Boithias formulation to improve agreement with measured results, especially at higher frequencies.
- H-Matrix would produce a large amount of files they have been compacted into one.
- Added a command-line utility for extracting complex impulse response data from binary output files.
- Significantly reduced APG runtime for large transmitter/receiver sets and increased overall CPU utilization during APG.

Bug Fixes

- Improved performance when loading large transmitter/receiver sets at the start of simulations.
- Fixed an issue in the API's GetChannelJob that caused an invalid channel index to be returned.
- Opening a project with unrecognized tags in batch mode will no longer cause the ui to hang.
- Allow GUI to import Odyssey/Planet files with updated antenna module.
- Fixed an issue where selecting a Tx/Rx/TxRx with some sets not shown would cause the incorrect item to be selected.
- The set_licenses option can take a space separated location.
- Increased the allowed facet geometry limited by stack.
- Viewing x paths of greatest power will no longer select ones of lowest power.
- Fixed an issue that caused X3D jobs using both APG and partitioning by transmitter point to produce invalid/corrupted results.

3.4.0.2

New Features

- Added an option to clean up database files from a study area's results directory and a corresponding application preference to do this automatically.

Bug Fixes

- Fix hanging edge check that incorrectly eliminated some edges and diffraction paths
- Prevent block of application when accessing the *Help* → *About Licensing* dialog.

3.4.0.1

New Features

- Added InteractionPointPrecision as an option in Project Properties.
- Updated the unrecognized keywords warning box to enable scrolling.

Bug Fixes

- Tighten tolerance for intersecting facets near another interaction.
- Fix memory leaks.
- Route layout sets will now find points at the same location as the calculation engine.
- Fix rounding of path power below -220 dBmW.
- Study area names now have white spaces trimmed.
- Correct bug where communications systems were unable to be properly renamed via GUI.
- Fixed minor bug that would not properly remove terrain cache file when replacing terrain via GUI.
- Fix bug where plots from binary sources would incorrectly display a missing file warning.
- Graphing bug fixed that would occasionally not load saved plots back in correctly.
- UAN improvements: 2-cut plane import fixed, handles sub 1-degree resolution correctly, better performance.
- Corrections to equation labellings in section 21.1 of the WI reference manual.
- Prevent crash due to plot references when saving project to a new location without copying study area results.
- User defined coefficient with spaces caused issues in running jobs, it is now fixed.
- Plots based on binary sources load correctly.

- Duplicating features multiple times gives new features unique short descriptions.
- Fixed an issue where a user defined file could be specified and copied into the project folder multiple times.
- Generated clearer error messages from X3D and allow simulations to continue if any recoverable error occurs.
- Fixed an issue that caused X3D simulations with MIMO antennas to crash when building Tx/Rx sets.

3.3.5.6

Bug Fixes

- Calculation log dialog now comes to the foreground when a calculation finishes as opposed to issuing a message.
- Fixed an issue where VSWR wasn't being proprly applied for loss calculations.
- Fixed an issue where VSWR and cable loss weren't correctly specified for each element in a MIMO antenna.
- Fixed an issue where plots made from output files not in a results folder would be incorrectly flagged as missing source files.

3.3.5.5

Bug Fixes

- Plots based on binary sources load correctly.
- Fixed an issue where UTM formatted Shapefiles would not import correctly if imported first.

3.3.5.4

Bug Fixes

- Fixed compilation issues with API example code.
- Fixed issue where facets coplanar with edge of scene could cause bad diffractions.
- Corrected diffraction calculation for interior angles.
- Fixed issue with Communication System Output not being available in the UI.
- Fixed an issue where the MIMO output browser was not accessible when the binary index file was accessible.

3.3.5.3

Bug Fixes

- Only update the Results tab when necessary to avoid UI hang.

- Correct bug where communications systems were unable to be properly renamed via GUI.
- Correct bug where duplicated Communication Systems were unable to be renamed.
- Fixed bug where plots from binary sources would incorrectly display a missing file warning.
- Fixed bug that would not properly remove terrain cache file when replacing terrain via GUI.

3.3.5.2

Bug Fixes

- Route layout sets will now find points at the same location as the calculation engine.

3.3.5.1

New Features

- The maximum application CPU thread count will by default be set to the number of CPU threads present in the system during installation.
- Added warning to Shapefile importer to inform when geolocation is unavailable.
- Added a warning when attempting to import and extrude a PolygonZ Shapefile, as the PolygonZ format shouldn't require extrusion.
- Added extent information and an attributes table to the Shapefile importer.
- Shapefiles in the UTM format will now attempt to set their zone automatically when selected for import.
- Extended region in which the Holm correction is applied to diffraction paths, to get better agreement with theoretical results.

Bug Fixes

- The application maximum CPU threads and study area CPU thread counts can no longer be specified as zero or negative values.
- The study area's geometry caching job will now use the the application's maximum number of threads to run.
- The Cartesian E-field phase components written to the exphs, eyphs and ezphs files are now correctly written in degrees.
- The transmitter input power written to the X3D .diag file is now correctly labeled as dBm.
- Importing shapefiles containing holes that intersect the exterior ring of a structure will no longer cause a crash.
- Removed crash caused by missing Shapefile projection file.
- Clarified warning message when Shapefile index or attribute database files are missing.

- Fixed an issue where specifying discrete colors in the project view's scale bar would incorrectly toggle checkboxes in the scale bar properties window.
- Fixed an issue where multi-file Shapefile imports wouldn't automatically set the coordinate system to import, and wouldn't list the entity type.
- Fixed an issue where changing a feature material when importing a Shapefile would cause a crash when exiting application or examining the feature's material properties.
- Changed handling of small minimum x-values so to ensure proper scaling when plotting.
- Fixed an issue where a crash could occur when viewing CDF plot properties or saving a project with a CDF plot present.
- Fixed an issue where a crash could occur when changing a graph's min and max values while a CDF plot is present.
- Fixed an issue where vertical surface receiver set indices could get out of sync when control points were edited.
- Fixed an issue where UTM Shapefiles could be misidentified and failed to properly set their zone.
- Fixed issue where UTM Shapefiles were incorrectly scaled when imported.
- Fixed an issue where imported UTM Shapefiles shift location upon reopen.
- Fixed an issue that could cause rays to leak through the intersection of one or more facets in the X3D model.
- Fixed an issue where attempting to import Shapefiles with geometry errors could result in a crash.
- Fixed an issue where paths were leaking through the intersection of an edge and a face.
- Fixed an issue where CPU thread count written to the .diag and .log files represented the user requested thread count and not the actual number of threads used.

3.3.5.0

Bug Fixes

- UAN importer works with radians and retains more digits of precision.
- Added validation error for missing antenna alignments.
- Will no longer overwrite existing diagnostic files if a licensing error is encountered or an invalid simulation file is specified. This prevents overwriting files from a previously successful run.
- Antenna patterns will now be colocated with the transceiver point at which they are activated for viewing in all cases.

3.3.4.2

Bug Fixes

- Geometry cache builder jobs will now properly report the number of CPU threads they are using in the calculation log.

3.3.4.1

New Features

- Stopped automatic creation of KMZ files in the project. Added ability to export KMZ/COLLADA files on a per-feature, or whole project, basis.

Bug Fixes

- Fixed an issue where the X3D model would attempt to run with no valid geometry; this includes either no geometry at all or only foliage geometry. The model will now issue an error message and exit when it cannot run due to invalid geometry.
- The maximum number of threads will now be properly accounted for when specifying study area thread counts.
- Plane wave power density values now have a wider valid range, and the user will be informed if the provided waveform and received power result in an unusable power density when editing a plane wave.
- .csv files exported by the MIMO Output Browser are now properly produced with comma delimiters.
- Disabled calculation of transmitters that are outside of the active study area.
- Updates to the UTD diffraction coefficients to improve reciprocity and better match measured cases for non-PEC wedge cases.

3.3.4.0

New Features

- The terrain pre-processing menu now contains an option to remove feature structures that lie completely outside the bounds of the terrain.
- Added the ability to align antennas in a given transceiver set to a common focal point. Also added spherical and nautical orientaion options for antennas.
- Disabled caching of FLEXlm licenses. Improved handling of licenses to make switching license sources easier.

Bug Fixes

- Fixed an issue where editing terrain, undoing changes and dismissing the window caused a crash.
- Fixed an issue where displayed antenna patterns did not persist when project was closed and reopened.
- Fixed an issue where transceivers sometimes display receiver control vectors for a transmitter point.
- Fixed an issue where UI would not render antenna rotations
- Fixed an issue with displaying control vectors without having to first view the antenna pattern.
- Fixed an issue where exporting a .uan file, would save a temporary .uan file in addition to the desired export.

- Fixed an issue where the UI was unable to load .uan files in an unsaved project.
- Added a check to directional and omni-directional antennas to ensure that the half-power beamwidth is less than the first null beamwidth.
- Fixed an issue where the UI was unable to process large geometry caching jobs in batch mode.
- Removed option to automatically center on selection and expose this operation only through the default view drop-down menu in the Project View.
- Multiple features imported from shapefiles are now properly saved.
- Fixed an issue that caused excessive runtimes when exporting H-matrix and other output types for MIMO simulations.
- Fixed an issue where exiting application before saving project would not prompt user to save.
- Timers now properly report running status. This was causing the calculation log window to never report a finishing time.
- Cropping features to terrain now properly handles features with a different origin than the terrain.
- Fixed an issue that prevented probability density plots from being generated.
- Fixed an issue when creating terrain profile graphs from Tx/Rx pairs.
- Geometry cache files are coppied to the new location during a Save As operation.
- Fixed an issue where XF MIMO antennas imported into an un-saved project were not saved.
- Output is flushed after a Save As operation.
- Plots that have lost connection to their underlying source files are now marked in the output tree. Plots that have lost connection to their source files can be more easily remedied.
- Renaming a Communication System no longer breaks existing plots or graphs. Deleting a study area or Communication System will remove all related plots from the project.

3.3.3.2

Bug Fixes

- Fixed bug where MIMO output browser failed to load data for transmitter sets with multiple points.

3.3.3.1

New Features

- Double clicking project files in an explorer window opens the project in a new instance of Wireless InSite or XGtd.

Bug Fixes

- Renaming a geometry feature's file now correctly renames the file on disk and saves the project.
- Replacing an in-use material no longer causes a crash.
- Plots missing unnamed files will no longer prompt the user to find the file at project open without a warning.
- Plots missing imported plot files will no longer prompt the user to find the file at project open without a warning.
- Prevented a crash when graphing the terrain profile. Terrain profile graphs are no longer empty.
- Output will refresh when its related scale bar is modified.
- Fixed issue where file names were incorrectly trimmed, causing data to be misplaced by the GUI.
- Fixed issue with saving projects containing plane waves. Plane waves may now be used in communications systems.
- Prevented a crash when deleting and replacing a material with one from the material database. Prevented a crash when deleting unknown layout transceivers.
- Prevented instances of duplicate licensing dialogs being displayed when loading a project by double clicking a .setup or .xgtd file in an explorer window.

3.3.3.0

New Features

- An exact interval may be specified for the terrain import sampling interval.
- Improved calculation engine logging when using EQI.
- Consolidated separate vertical and horizontal foliage attenuation values into a single dB/m value.
- Improved foliage geometry processing to allow for 3D foliage instead of 2D footprints extruded down to the ground. This also includes support for raised canopies in X3D.

Bug Fixes

- Improved experience loading and using user-defined antennas: GUI will only load UAN when data is required. Also eliminated unnecessary re-rendering.
- Fixed an issue that caused binary output files over 4 GB in size to be unreadable in the Output Browser.
- Fixed an issue that caused a crash when using EQI and partitioning by transmitter point.
- Removed erroneous popup when attempting to crop foliage to a study area boundary when there is no terrain.
- Improved stability of conforming foliage to terrain when performed multiple times.
- Reduced the number of facets generated when conforming foliage to terrain.

- Improved process for changing foliage heights. If the foliage height is raised, then its thickness is retained and a bottom canopy is added to make the foliage a closed geometry. If the foliage is lowered so that its previous lower height falls below the terrain then it will no longer have a bottom canopy. If the foliage has a bottom canopy and is raised or lowered so that the bottom still lies above the terrain, then the foliage is translated with no structural modifications.
- Duplication of grouped items also duplicates the material assignment. Prior to this change, lack of a material assignment caused the sub-items not to be rendered and could lead to a crash.
- Change foliage height now works correctly at any group level.
- Movie player now shows animated fields as a display option for models that support that output type (Full3D, Urban Canyon, Vertical Plane).
- Improved wording of click and drag help message.
- User Defined materials will now use correct reflection and transmission coefficients.

3.3.2.0

Bug Fixes

- Removed a run mode that was only available in some legacy propagation models, and could lead to erroneous results: 'Changed heights (Tx/Rx/City/etc)'. This run mode was never available when using X3D.
- Improved automatically-generated titles in point-to-point output plots and graphs. They now include output type and transmitter ID.
- Improved the experience of multiple users on a single machine. Default registry entries will be created if they cannot be found.
- Temporary projects will not appear in the recent projects list.
- Deletion of an in-use antenna allows for replacement with an identical antenna. This corrects a situation that could lead to a UI crash.
- Eliminated a case of an unnecessary prompt to save a project. A project is no longer marked out-of-date by viewing (but not changing) antenna properties.
- GUI no longer crashes when saving preferences after closing a project.
- Graphs using point-to-point data scale consistently between unit changes and opening and closing projects.

3.3.1.1

Bug Fixes

- Running a project with a Communication System will no longer invalidate the geometry cache.

3.3.1.0

New Features

- When requested, X3D API can create a pre-processed geometry cache for later reuse, thereby avoiding geometry processing time for jobs when the geometry has not changed. The API user must regenerate the cache when geometry has changed. The GUI will automatically manage this request and indicate to the user when a cache is available for subsequent runs.
- Significantly reduced geometry preprocessing times for the X3D model.
- Significant speed-up for geometry preprocessing for all GPU-based X3D model runs.
- Optimization for pre-processing diffraction edges in scene geometry.
- Software version now appears in the main application window's title bar.
- X3D foliage and additional GDAL format options are enabled by default.
- Added units to the dielectric material properties window.
- Study areas that haven't been modified no longer make the project out of date.
- Updated the GUI to maintain the cached results of geometry processing and reuse it for subsequent simulations until the geometry in the project is modified and saved.
- Binary output files contain TxRx short descriptions in their headers.
- Added a window to display the full license info file while a project is open.
- X3D's calculation log output and diagnostic file are now more concise.

Bug Fixes

- Exporting antenna patterns using rotations now matches how antennas are rotated for a transceiver set.
- Transmission phase will now better account for passage through the material.
- Materials with permittivity less than 1, are now treated as having permittivity of 1.
- Changing the description of a study area or the project filename will no longer prevent related plots from loading properly.
- Now able to select a different tabulated beam forming file with the same name.
- GDAL terrain importer will not crash when attempting to read index files.
- Imported terrain is saved with its short description as the file name.
- Deleting simulation results no longer breaks file connections for plots and graphs.
- Corrected many issues related to antenna plotting. Antenna plots now behave similar to output plots in that if the underlying antenna is modified they will
- When plotting an antenna pattern a duplicate of the graph that is not added to the graph list is no longer created.
- Stopped geometry cache from regenerating for disabled study areas.

- Adjusted progress reporting during X3D ray-tracing.
- Facets which are too small to be used by the X3D model will be automatically removed from the scene.
- Prevented crash when removing MIMO antenna's sub-antennas.
- Plots properly scale axis when changing unit preferences. Log(distance) plots will not fail to plot with very large or very small values.
- Imported UAN files show the correct maximum gain value.

3.3.0.4

Bug Fixes

- User-defined antennas in MIMO arrays can set their user-specified max gain.
- The previously-removed X3D CPU mode extension no longer interferes with requesting a GPU for X3D study areas.
- RSSI is now calculated using the interference with its strongest transmitter.
- Excess path loss, freespace path loss, and freespace power are now generated when diffuse scattering is enabled.
- Directional antennas calculated with RXAPI use the E-plane first null beamwidth and H-plane first null beamwidth parameters correctly.

3.3.0.2

Bug Fixes

- Noise power density field is enabled when calculating interference and noise. SISO interference and noise calculation properly scales the noise power density by the signal bandwidth. MIMO interference and noise calculation now uses the signal bandwidth.
- Resolved issue causing crash during Odyssey/Planet antenna imports.

3.3.0.1

Bug Fixes

- MIMO antenna elements update their short descriptions when the parent antenna's short description changes.

3.3.0.0

New Features

- Added new MIMO capabilities and outputs to the Communications Systems Analyzer, including MIMO beamforming, diversity, and spatial multiplexing. Includes new user-defined precoding table input, as well as several new outputs: RMS delay spread, angle spreads of arrival and departure, and element-to-element path loss. Also added handling of VSWR and cable loss for MIMO systems. Changes incorporated SINR, throughput analysis, and BER analysis for MIMO systems as well.
- Binary Output CSV files now have descriptive headers.
- MIMO transmitters can now set power on a total array basis, or a per element basis.
- X3D calculations now include the effects of transmission line loss.
- MIMO outputs now include element-to-element path loss.
- Added Do-Not-Extrude checkbox for shapefile import, to support importing with different heights to simulate floors in a building.
- Tx and Rx short descriptions are now present in X3D output files.
- Added -licenses command line option to wibatch and wfbatch and xgbatch to display a list of available licensed features.
- MIMO Path loss and path gain can now be viewed and plotted under received power in the output browser.
- Communication system strongest transmitter files now correctly display the distance between the receiver point and its corresponding strongest transmitter. Distance is now displayed as 0 meters for receiver summary files for non-route receiver sets.

Bug Fixes

- TxRx Trajectory set 'Include Roll Angle' setting is now preserved when re-opening existing projects.
- Improved normalization of User-Defined antenna patterns.
- Fermi-based GPUs, and out of date GPU drivers no longer cause crash.
- Wireless access method (WAM) tables are no longer incorrectly reported as being invalid.
- Noise values are now correct for receiver summary statistics calculations in the communication systems toolbox.
- BER and MIMO-SVD jobs will properly test if the Matlab libraries are available before running.
- MIMO output files are no longer generated for X3D study areas that do not contain any active MIMO antennas.
- Loading Communication systems projects created prior to 3.3 now correctly preserves interference between base stations settings.
- WAM files are now moved with the project when the Save As function is used.
- Communication systems using precoding tables now correctly determine how many elements should be used to verify that the table matches the number of active base stations.

- Fixed memory leak that could occur when switching antennas.
- Power, RSSI, and RSRP quantities in receiver summary and receiver summary LTE files are now correctly presented in dBmW instead of dBW.
- Total power with phase is now correctly calculated for receiver summary and receiver summary LTE files.
- Corrected default values for Signal to Interferer Ratio.
- Corrected SINR calculation when using MRT.
- Corrected SINR calculation when precoding.
- Changes to uniform interference are now properly handled.
- MIMO Array Builder defaults to Wavelengths of 0.5 when building a MIMO array.
- Added warnings when a user-defined throughput fill is specified but has been removed from the project folder.
- Imported XF MIMO antennas have gain values after import.

3.2.10.0

Bug Fixes

- Improved ray tracing performance in cases with large, dense receiver sets.

3.2.9.0

Bug Fixes

- Imported features now have appropriate file extensions after renaming.

3.2.8.0

New Features

- Added phase to the X3D, VPUP, and TPG Path output file.
- Multiple geometry features can be imported at the same time.
- All X3D, VPUP, and TPG output files contain headers with descriptive units.
- FLEXlm 9 dongle keys are reported in the license log.

Bug Fixes

- MIMO antennas imported from XF will no longer overwrite user defined antenna files.
- Complex E-Field output file now has proper units in header.
- Propagation along hanging edges is no longer affected by the normal direction of the facet.
- Imported Odyssey and Planet antennas are now given proper filenames.
- Walfisch Ikegami model calculations for Building Shadowing Loss and Roof-to-Street Loss have been corrected.
- APG will now run with active Transceivers sets.

3.2.7.0

New Features

- X3D log file now shows the run times for individual steps of its calculation.

Bug Fixes

- Viewing results in the MIMO output browser will no longer cause a crash.

3.2.5.0

New Features

- On Windows, updated support for NVIDIA CUDA micro-architectures Kepler (3.x), Maxwell (5.x), and Pascal (6.x). Fermi (2.x) cards are no longer supported.

Bug Fixes

- Antenna patterns no longer disappear when examined.

3.2.4.0

New Features

- MIMO antennas are now created with the default antenna automatically added. New antennas can be added by double-clicking on empty rows of the antenna table.
- A terrain that has been cropped by a study area will now remain cropped after saving.

Bug Fixes

- Binary output for MIMO simulations is now always correctly written when loading projects created in an older version.
- Corrected error with probability density and cumulative distribution plotting the phi angles for mean direction of arrival and mean direction of departure when the theta angle was requested.
- Plotting values versus distance to a vertical surface receiver set now uses the distance from the transmitter point.
- Features will now only prepare for rendering if they are visible.
- Selecting a feature or face that is colored by height will now highlight that selection.
- An additional default antenna is no longer added to existing MIMO antennas.

3.2.0.6

Bug Fixes

- Shapefile, STL, and Terrain imports are now given the correct short description.
- Terrain imports no longer crash when an invalid file is selected.

3.2.0.5

New Features

- Improved importing of shapefiles and added support for shapefiles containing geometry with holes (such as courtyards).

3.2.0.4

New Features

- User defined Antenna, Waveform, and Material files automatically rename themselves when imported with conflicting names. Added menu option for removing unnecessary project files.
- Improved handling of feature files with duplicate names.

Bug Fixes

- User-defined antenna files assigned to a MIMO array are no longer removed during project folder cleaning.
- MIMO antennas using Patch antennas will no longer fail.
- Use of original calcuation engine to calculate antenna patterns will no longer fail.
- Viewed antenna pattern will update when changing antenna properties.

- Shapefiles without projection files no longer stretch after importing. Users should be aware that a shapefile with a projection file produces the best import results.
- Fixed crash that occurred shortly after attempting to import certain shapefiles.
- Loading a feature file with associated user defined material files from a location outside of the current project directory, now copies the user defined files on Save or Save As.

3.2.0.3

Bug Fixes

- Features loaded from another project folder now save in the folder of the current project.
- Feature files are no longer duplicated when using the Save As function.
- Features loaded from another project folder that contain user defined materials will now copy related material files upon saving.
- Fixed bug in header of .rsumlte files, that caused Total Power with Phase to not be plottable or viewable.

3.2.0.2

Bug Fixes

- Fixed occasional crash that occurred with the combination of MIMO, diffuse scattering, and foliage.
- Overlapping co-planar faces no longer allow diffraction paths to leak through the facet-edge interface.

3.2.0.1

New Features

- Study area property windows now show default parameter values instead of 'Auto' for applicable model types.

Bug Fixes

- Propagation paths in the X3D model no longer diffract off of edges consisting of three or more facets.
- Running a communication system analysis will no longer crash after deleting a transmitter set.
- Communication systems no longer allowed to have 'mhz' appear in their short description which caused file look-up collisions with other output types.
- Communication systems enforced to have unique short descriptions to prevent them from overwriting each other's files. Each system writes its files to a folder matching this value.

3.2.0.0

New Features

- Upgrades to Wireless InSite Communications System Toolbox to support Throughput for Wi-Fi protocols 802.11n and 802.11c. This supports the calculation of the throughput for single-input-single-output (SISO) systems, with an optional multi-stream parameter to provide an initial estimate of potential Multiple-Input-Multiple-Output (MIMO) performance.
- Wireless InSite Communications System Toolbox upgraded to support Shannon capacity calculations. This output represents the theoretical upper bound on the data rate of a channel, either for a specific protocol or for a generic communications channel, based on the Signal-to-Interference-plus-Noise Ratio (SINR) and the bandwidth of the channel.
- Wireless InSite Communications System Toolbox outputs expanded to include new communications metrics: Signal-to-Interference-plus-Noise (SINR), the Received Signal Strength Indicator (RSSI), and two standard LTE metrics: the Reference Signal Received Power (RSRP) and the Reference Signal Received Quality (RSRQ).
- Wireless InSite Communications System Toolbox Bit Error Rate (BER) calculations expanded to support higher order modulation coding schemes, including 256 QAM, 512 QAM, and 1024 QAM, allowing analysis of BER for communications channels relevant to current and future planned protocols.
- Wireless InSite Communications System Toolbox now provides a user-defined throughput capability that allows users to specify their own mapping between the SINR and the throughput for specified bandwidths, providing a mechanism for representing vendor-specific, or experimental mappings between signal quality and modulation coding scheme.

3.1.5.0

New Features

- X3D and Real Time models now run when more than 128 materials are present. Duplicated materials now save properly.
- Solid body and 3D are now the default views in the Project View window.

Bug Fixes

- Tx/Rx antenna pattern visibility is now preserved between sessions.
- Tx/Rx Sphere set spacing is now properly labeled in degrees.
- Duplicating X3D study areas now duplicates Queuing and Partitioning appropriately.
- GUI no longer hangs or crashes when changing a large number of materials in one or more features.
- Saving the project no longer prompts the saving of individually selected structures.
- wibatch script now supports command line parameters with spaces.

3.1.4.0

New Features

- Added support for user-defined materials to the X3D model.

3.1.2.0

Bug Fixes

- Project and application preference settings are now preserved from one version to the next.
- Fixed erroneous diffraction edges caused by precision on small facets.
- Fixed a bug which caused modified atmosphere properties not to be saved after X3D study area creation.

3.1.1.0

New Features

- Optionally import description for user-defined Tx/Rx sets.

3.1.0.1

Bug Fixes

- Fixes an issue where, in some diffraction cases, power was not reciprocal on individual paths.

3.1.0.0

New Features

- Modified ordering of Project Hierarchy window to match Main Window tabs.

Bug Fixes

- Antennas with array information are now read from the project file correctly.
- Do not adjust the element rendering size in the MIMO antenna properties window unless a valid value is specified.
- Issue error when building a MIMO antenna array with invalid values for number of elements or spacing.

- Correction to mean direction of departure and arrival calculations used by the Full 3D, Vertical Plane, and Urban Canyon propagation models.
- An imported plot will use the name of the imported file as its description in the graphs legend.
- The distances listed in output files for receiver locations in point and polygon sets are now calculated from the transmitter location.
- Will no longer close current project without allowing user to save changes when opening another project.

3.0.2.0

New Features

- Commercial release of Wireless InSite Diffuse Scattering capability.

Bug Fixes

- Properly handle exact path verification in MIMO for diffuse scattering Paths.

3.0.1.0

Bug Fixes

- Added power weighting to mean direction of arrival/departure calculations.
- Improved interpolation method for UAN antennas that specify phase.

3.0.0.10

New Features

- Replaced transmitter set partitioning with transmitter point partitioning for X3D study areas. Older projects that specified transmitter set partitioning will change silently to point partitioning. Previous software versions disable partitioning if transmitter point partitioning is specified.

3.0.0.9

Bug Fixes

- Antenna patterns can be graphed and will render correctly in properties window and Project View when activated for a transceiver set.
- Project View no longer clips far zone gain and transceiver antenna patterns.

3.0.0.8

New Features

- Added PEC-Backed material support to X3D calculation engine.

3.0.0.7

New Features

- Added diffuse scattering final interaction only flag to X3D calculation engine and GUI.
- Added LayeredDielectric Support to X3D calculation engine.

Bug Fixes

- Modified text in diffuse scattering properties dialog for better clarity.
- When a study area is deleted remove related views in the MIMO output browser.

3.0.0.5

New Features

- Minor speedup in X3D geometry processing in many cases.
- No longer render walls of floor plan features using material thickness.

Bug Fixes

- Add checkbox next to ‘Output’ selection mode when it is active.
- No longer copy original KMZ or COLLADA asset into the project directory as a part of the import process.
- Features created using the editors in the application are no longer allowed to have the same file names as other features.
- Opening a pre 3.0 project file no longer requires user to re-save.
- Opening a feature file into a project that has a conflict with the file name of another feature will now write itself out to a seperate file when saving the project.
- When opening a new project changes made to the current project are saved to the correct project folder.
- When features are removed from the project using the ‘Crop geometry to’ option on a study area, references to their materials are removed from the material table.
- When a material is removed from a feature the project hierarchy is now updated to remove its reference.

3.0.0.4

Bug Fixes

- Corrected file names for MIMO DoA/DoD files.

3.0.0.3

New Features

- Project View now displays an orientation triad in the lower left corner. This indicates the current rotations the view applies to the global coordinate system when displaying the project.

Bug Fixes

- Fixed issues with how maximum gain in UAN file is handled. Expanded user options for setting the maximum gain in the user-defined antenna properties window.
- Corrected issue with the serialization of parameters for several material types.
- Corrected an issue with Freespace, Hata and COST-Hata sometimes running twice and sometimes producing incorrect or empty output files.

3.0.0.1

New Features

- Commercial release of Wireless InSite MIMO capability, including MIMO Array builder, MIMO Output browser, XF MIMO Array import, and optimized MIMO simulation capability.
- Commercial release of Wireless InSite's Adjacent Path Generation (APG) optimization for simulations involving large or dense receiver sets.

Bug Fixes

- Opening project files before output requests lists were maintained by individual models will now apply the global request list of the project. This ensures that output is produced as expected based on previous settings.

3.0.0.0

New Features

- Added material properties to database for wi-fi and millimeter wave frequencies based on ITU recommendations.

- KMZ extended to be default file format for all feature types.
- When features are imported their file names are set according to the source file name. If the same source file is used multiple times, each feature is given a unique filename by appending a duplicate number to it. The features short description is also set based on the file name for further clarification.
- Added ability to crop foliage to a study area's boundary. Consolidated this with the 'Crop terrain to boundary ' and 'Remove extraneous buildings outside' operations into a single 'Crop geometry to' option in the context menu.
- Added MIMO Tutorial to User's Guide.

Bug Fixes

- Corrected backwards compatibility issue when reading the output request list from a project that does not have lists at the model level.
- The output browser now correctly handles projects with spaces in their names.
- Corrected issues with the saving of requests at the project level which is reserved for use by calcprop.
- Fixed a repeatability issue with input power in Monte Carlo calculations.